

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-VI EXAMINATION – WINTER 2025

Subject Code:3160715

Date:25-11-2025

Subject Name:System Software

Time:02:30 PM TO 05:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	MARKS
Q.1 (a) Define the following terms:	03
i) System Programming	
ii) Logical Address Space	
iii) Device Driver	
(b) Discuss the various recent trends in software development.	04
(c) Explain the role of symbol table data structure in language processor and also, discuss how one can organize the symbol table using linear data structure?	07
Q.2 (a) Discuss various desirable properties of an intermediate representation (IR) of a code in language processor.	03
(b) Explain the following assembler directives:	04
i) END	
ii) ORIGIN	
(c) An assembly program contains the statement A EQU B+10 Indicate how the EQU statement can be processed if	07
i) B is a back reference	
ii) B is a forward reference	
OR	
(c) Discuss in detail the various data structures involved in PASS-I of the assembler.	07
Q.3 (a) What do you mean by lexical and semantic macro expansion?	03
(b) Explain nested macro call with example.	04
(c) Compare and contrast the properties of macros and subroutines with respect to the following:	07
i) code space requirements	
ii) execution speed	
iii) processing required by the assembler	

OR

Q.3 (a) Draw the schematic design of a macro preprocessor.	03
(b) Explain the use of REPT and IRP statement in macros.	04

- (c) Explain the various attributes of the following tables involved in macro preprocessor. **07**
- i) MNT
 - ii) KPDTAB
 - iii) SSTAB
 - iv) EVTAB
- Q.4 (a)** What do you mean by non relocatable program? **03**
- (b)** Explain absolute loader with suitable example. **04**
- (c)** Discuss in detail how linker resolves the external references. **07**
- OR**
- Q.4 (a)** Define the following terms. **03**
- i) Overlay program
 - ii) Relocating loader
 - iii) Program relocation
- (b)** A self-relocating program needs to find its load address before it can execute its relocating logic. Comment on how this information can be determined by the program. **04**
- (c)** Explain various components of an object module. **07**
- Q.5 (a)** Describe the ambiguity in grammar specification. **03**
- (b)** Explain the difference between pure and impure interpreter. **04**
- (c)** Demonstrate the use of triples and quadruples representation used in compiler with suitable example. **07**
- OR**
- Q.5 (a)** Explain the role of LEX and YACC utility. **03**
- (b)** Discuss the top down parsing without backtracking with example. **04**
- (c)** Discuss various optimizing transformation techniques commonly used in compilers. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI EXAMINATION – SUMMER 2025****Subject Code: 3160715****Date: 30-05-2025****Subject Name: System Software****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Define following: 1)Specification Gap 2)Language Processor 3)Language Migrator	03
	(b) Differentiate Problem Oriented Languages and Procedure Oriented Languages.	04
	(c) Describe System Software Development with appropriate diagram.	07
Q.2	(a) Define Left Recursion. Remove Left Recursion from the following grammar: $A \rightarrow A x \mid A y \mid A B \mid c \mid d$ $B \rightarrow e$	03
	(b) Construct LL(1) Parsing table for the following grammar: $S \rightarrow aAC \mid Bb$ $A \rightarrow eD$ $B \rightarrow f \mid g$ $C \rightarrow h \mid i$ $D \rightarrow bE \mid \epsilon$ $E \rightarrow eD \mid dD$	04
	(c) Explain advanced assembler directives with suitable example.	07
	OR	
	(c) Consider following assembly language program: (i) Write equivalent machine code (ii) Write Intermediate code using Variant-I representation. START 200 READ A READ B MOVER BREG, A MULT BREG, B MOVEM BREG, C STOP A DS 1 B DS 1 C DS 1 END	07
Q.3	(a) Write a macro with parameters A, B, C and D. Calculate $A * B + C * D$ in AREG.	03
	(b) Explain nested macro call with example.	04

	(c)	Explain expansion time loops in macro.	07
		OR	
Q.3	(a)	Compare features of subroutine and macro.	03
	(b)	Explain positional parameter, keyword parameter and default value parameter.	04
	(c)	Explain expansion time statements AIF, AGO and ANOP.	07
Q.4	(a)	Compare loader and linker.	03
	(b)	Explain Bootstrap loader.	04
	(c)	What is Overlay? Explain the execution of an overlay structured program.	07
		OR	
Q.4	(a)	Explain Compile-and-Go loader.	03
	(b)	Consider an object module with three programs A, B and C, located at following addresses: Program Address A 200-250 B 302-370 C 480-533 The load address is 300 for the executable A, B and C to be generated. Find: i) Size of each Program ii) Load origin of each program iii) Relocation Factor for each Program	04
	(c)	Explain self-relocating program.	07
Q.5	(a)	Explain analysis phase of an assembler.	03
	(b)	Write quadruple representation for the following expression: Result = -w * x + -y * z	04
	(c)	Explain pure and impure interpreters.	07
		OR	
Q.5	(a)	Compare one pass assembler and two pass assemblers.	03
	(b)	Describe static binding and dynamic binding.	04
	(c)	Describe JVM with appropriate diagram.	07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-VI (NEW) EXAMINATION – WINTER 2024****Subject Code:3160715****Date:05-12-2024****Subject Name:System Software****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Explain semantic gap, specification gap and execution gap.	03
	(b) Explain the various tasks performed during analysis and synthesis phases of the assembler.	04
	(c) List various phases of Language Processor. Explain any one phase in detail.	07
Q.2	(a) Describe ambiguous grammar with suitable example.	03
	(b) Explain Left recursion in brief and remove left recursion from the following grammar: S -> A A -> aB / Ad B -> bBC / f C -> g	04
	(c) Explain various Assembler directives with suitable example.	07
	OR	
	(c) Describe following data structures: OPTAB, SYMTAB, LITAB and POOLTAB.	07
Q.3	(a) Define: Parse tree, Syntax tree, DAG	03
	(b) Differentiate between single pass Assembler and 2-pass Assembler.	04
	(c) Explain recursive descent parsing algorithm with example.	07
	OR	
Q.3	(a) Define: Token, Lexeme and Pattern	03
	(b) Construct LL(1) parsing table for following grammar. S -> iCtSeS iCtS a C -> b	04
	(c) Obtain minimized DFA for the following regular expression: aa*bc*#	07
Q.4	(a) Explain handle pruning with respect to the bottom up parsing.	03
	(b) Explain macro preprocessor in brief.	04
	(c) What is intermediate code? Explain types of intermediate code in compilers.	07
	OR	
Q.4	(a) Define: Subroutine, DEFTAB, ARG TAB	03
	(b) Explain basic structure of a LEX program.	04
	(c) Define macro in the context of programming language. Explain the structure of a macro definition.	07

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|------------|------------|--|-----------|
| Q.5 | (a) | Enlist different loading schemes. | 03 |
| | (b) | Describe Absolute loader in brief. | 04 |
| | (c) | Explain how relocation and linking is performed. | 07 |
| | | OR | |
| Q.5 | (a) | Enlist the languages using interpreter. | 03 |
| | (b) | Describe the classification of Debuggers. | 04 |
| | (c) | Define: Interpreter, Explain the merits and demerits of interpretation | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2024****Subject Code:3160715****Date:24-05-2024****Subject Name: System Software****Time: 10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Explain fundamental of language processing.	03
	(b) Define following terms:	04
	1) Language Migrator	
	2) Execution gap	
	3) Token	
	4) Handle	
	(c) Explain with examples - expansion time variables, expansion time Statements -AIF and AGO for macro programming.	07
Q.2	(a) Apply rule to remove left recursion from following grammar.	03
	$S \rightarrow A$	
	$A \rightarrow Ad \mid Ae \mid aB \mid aC$	
	$B \rightarrow bBC \mid f$	
	$C \rightarrow g$	
	(b) Construct LL(1) parsing table for following grammar.	04
	$S \rightarrow iCtSeS \mid iCtS \mid a$	
	$C \rightarrow b$	
	(c) Describe following data structures: OPTAB, SYMTAB, LITTAB and POOLTAB	07
	OR	
	(c) Explain and compare two variants of the intermediate code generated from multi pass assembler.	07
Q.3	(a) Compare top-down and bottom-up parser.	03
	(b) Explain inherited and synthesized attributes in detail with example.	04
	(c) Consider following assembly language program: Create equivalent machine code	07

	START	101
	READ	N
	MOVER	BREG, ONE
	MOVEM	BREG, TERM
	MULT	BREG, TERM
	MOVER	CREG, TERM
	ADD	CREG, ONE
	MOVEM	CREG, TERM
	COMP	CREG, N
	BC	LE, AGAIN
	MOVEM	BREG, AGAIN
	PRINT	RESULT
	STOP	
N	DS	1
RESULT	DS	1
ONE	DC	'1'
TERM	DS	1
	END	

Instruction opcode: STOP – 00, ADD – 01, MULT – 03, MOVER – 04,

MOVEM – 05, COMP – 06, BC – 07, READ – 09, PRINT – 10, LE – 02

Assembler directives: START – 01, END – 02

Declaration statements: DC – 01, DS – 02

Register code: BREG – 02, CREG – 03

OR

- Q.3 (a)** Compare and Contrast macro preprocessor and macro assembler. **03**
- (b)** Explain use and field of following tables of a macro KPDTAB, MDT, EVTAB, SSTAB **04**
- (c)** Consider following assembly language program: Create equivalent machine code **07**

```

START      300
MOVER      AREG, '=5'
MOVEM      AREG,R1
AGAIN      MOVER      AREG,R1
           MOVER      CREG,R2
           ADD        CREG,='1'
           BC         ANY,TERM
           .....
           LTORG
           = '5'
           = '1'

TERM       SUB        AREG,='1'
           BC         LT,DOWN
LAST       STOP
           ORIGIN     LOOP+2
           MULT       CREG,R2
           ORIGIN     LAST+1
R1         DS         1
DOWN       EQU        LOOP
R2         DS         1
           END
           = '1'

```

- Q.4 (a)** Explain DAG with example. **03**
- (b)** Explain design of a linker by addressing issues of relocation and linking. **04**
- (c)** Given a grammar, **07**

$E \rightarrow TA,$
 $A \rightarrow +TA \mid \varepsilon$
 $T \rightarrow VB$
 $B \rightarrow *VB \mid \varepsilon$
 $V \rightarrow id \mid (E)$

Develop an LL (1) parser table and parse following string using the parsing table. id * (id + id)

OR

- Q.4** (a) Discuss parameters for Activation Records **03**
 (b) Explain characteristics of self-relocating programs. **04**
 (c) Define Operator precedence grammar. Convert following production rules of grammar into suitable Operator precedence grammar. **07**
 $E \rightarrow EAE \mid id$
 $A \rightarrow - \mid *$
- Q.5** (a) Explain Left factoring with example. **03**
 (b) List out various Code Optimization techniques used in Compiler. Explain any two techniques with suitable example. **04**
 (c) Generate Quadruple, Triple, Indirect Triple for following expression: $ans = a + b * c / 2.0$ **07**
- OR**
- Q.5** (a) Explain the terms Binding and Binding Times. **03**
 (b) Explain pure and impure interpreters. **04**
 (c) What is Symbol table? Explain how one can organize Symbol table using Linear Data Structure? **07**
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GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – WINTER 2023****Subject Code:3160715****Date:13-12-2023****Subject Name: System Software****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Explain different kinds of system software.	03
	(b) Difference between Procedure oriented language and Problem oriented language.	04
	(c) Explain the various stages of the life cycle of a source program with a neat diagram.	07
Q.2	(a) Eliminate left recursion $E \rightarrow E + T \mid T$	03
	(b) Describe following data structures: OPTAB, SYMTAB, LITAB and POOLTAB.	04
	(c) Compare Top Down and Bottom Up Parsing. Find out First and Follow and construct LL (1) parser table for following Grammar: $E \rightarrow TE'$ $E' \rightarrow +TE' \mid \epsilon$ $T \rightarrow FT'$ $T' \rightarrow *FT' \mid \epsilon$ $F \rightarrow (E) \mid id$	07

OR

- (c)** Consider following assembly language program: Show (i) Contents of Symbol Table (ii) Intermediate codes using Variant-I representation. **07**

```

START 101
READ N
MOVER BREG, ONE
MOVEM BREG, TERM
AGAIN MULT BREG, TERM
MOVER CREG, TERM
ADD CREG, ONE
MOVEM CREG, TERM
COMP CREG, N
BC LE, AGAIN
MOVEM BREG, AGAIN
PRINT RESULT
STOP
N DS 1
RESULT DS 1
ONE DC '1'
TERM DS 1
END

```

Instruction opcode: STOP – 00, ADD – 01, MULT – 03, MOVER – 04, MOVEM – 05, COMP – 06, BC – 07, READ – 09, PRINT – 10, LE – 02

Assembler directives: START – 01, END – 02

Declaration statements: DC – 01, DS – 02

Register code: BREG – 02, CREG – 03

- Q.3** (a) Draw a flowchart of maintaining Table of Incomplete Instruction (TII) in assembler. **03**
- (b) Explain REPT and IRP statement with proper example. **04**
- (c) Explain advanced assembler directives with suitable example. **07**
- OR**
- Q.3** (a) Explain memory allocation in block structured language. **03**
- (b) What is the difference between Keyword parameters and positional parameters? **04**
- (c) Explain lexical and semantic expansion of macro with example. **07**
- Q.4** (a) Write regular expressions of a given language. The language consists no. of zero should be multiple of 3 **03**
- (b) Explain in brief design of an absolute Loader. **04**
- (c) Explain Self relocating program and overlay structure program. **07**
- OR**
- Q.4** (a) Write a regular expression for a language containing both 11 and 00 as substring **03**
- (b) Write and explain the algorithm for macro expansion. **04**
- (c) What is program relocation? How relocation is performed by linker? Explain with example. **07**
- Q.5** (a) Explain types of grammar. **03**
- (b) Explain the front end of toy compiler with suitable example. **04**
- (c) What is interpreter? Explain pure & impure interpreters. **07**
- OR**
- Q.5** (a) Explain Bootstrap loader. **03**
- (b) Given following expression: $x = -a * b + -a * b$ **04**
- (1) Write three address codes for the expression.
- (2) Give triple implementation for the three-address code of the expression.
- (c) What is interpreter? Explain benefits of interpreter. Compare interpreter and compiler. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2023****Subject Code:3160715****Date:14-07-2023****Subject Name:System Software****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

			MARKS
Q.1	(a)	Compare user-centric view and system-centric view of system software.	03
	(b)	Enlist levels of System Software. Explain any two in detail.	04
	(c)	Define Language Processing. List various phases of Language Processor. Explain each phase in detail.	07
Q.2	(a)	Explain the causes of Large Semantic Gap.	03
	(b)	An assembly program contains the statement X EQU Y+25 Indicate how the EQU statement can be processed if a) Y is a back reference b) Y is a forward reference.	04
	(c)	Given the source program:	07
		START A DS L1 MOVER ADD MOVEM D EQU L2 PRINT ORIGIN C DC ORIGIN STOP B DC END 100 3 AREG, B AREG, C AREG, D A+1 D A-1 '5' L2+1 '19' L1	
		a) Show the contents of the symbol table at the end of Pass I.	
		b) Explain the significance of EQU and ORIGIN statement in the program and explain how they are processed by the assembler.	
		c) Show the intermediate code generated for the program.	

OR

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|---|-----------|
| (c) Differentiate one pass and two pass assembler. Explain how forward references are handled in two pass assembler. | 07 |
|---|-----------|

- Q.3 (a)** Compare and contrast the properties of macros and subroutines with respect to following: **03**
1. Code space requirements
 2. Execution speed
- (b)** Explain use of AIF and AGO with example. **04**
- (c)** Define a macro taking starting_location and N as parameters to find summation of all N numbers stored at location starting from starting_location. The result is to be stored at starting_location. **07**
- OR**
- Q.3 (a)** Explain the use of expansion time loop. **03**
- (b)** Explain macro definition and call in detail. **04**
- (c)** Define a Macro taking A and B as parameters to compute $A=A*B+B*B+A*B$. **07**
- Q.4 (a)** Explain linking of overlay structured program. **03**
- (b)** Explain Compile-and-Go loaders with example. **04**
- (c)** Write and explain an algorithm for first pass of a linker. **07**
- OR**
- Q.4 (a)** Explain absolute loader with suitable example. **03**
- (b)** Draw the flow chart for the dynamic linking. **04**
- (c)** With algorithm and example, explain how relocation is performed by linker? **07**
- Q.5 (a)** Discuss dead code elimination method with suitable example. **03**
- (b)** Explain JVM in detail. **04**
- (c)** Define Simple Phrase and Handle. Using Handle and Simple Phrase trace the bottom up parsing algorithm. **07**
- Grammar is :
- $E \rightarrow T + E \mid T - E \mid T$
- $T \rightarrow T * V \mid T / V \mid V$
- $V \rightarrow a \mid b \mid c \mid d$
- String is : $a - b * c + d$
- OR**
- Q.5 (a)** Discuss the limitations of stack based memory allocation. **03**
- (b)** Explain classification of debuggers. **04**
- (c)** Construct optimized DFA for following Regular Expression. **07**
- $(1^*)^* 0(0|1)^* \#$

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI(NEW) EXAMINATION – WINTER 2022****Subject Code:3160715****Date:17-12-2022****Subject Name:System Software****Time:02:30 PM TO 05:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

Q.1	(a) Differentiate Compiler and Interpreter.	03
	(b) Define following terms: 1)Linker 2)Loader 3) Parser 4)Assembler	04
	(c) Explain phases of Compiler with suitable example.	07
Q.2	(a) What is a Symbol Table?	03
	(b) Define following terms: 1)Semantic Gap 2)Execution Gap	04
	(c) Find First and Follow Set from following grammar S-> Aa bAc Bc bBa A-> d B-> d Check grammar is LL(1) or not?	07
	OR	
	(c) Find First and Follow Set from following grammar S-> AaAb BbBa A-> € B-> € Check grammar is LL(1) or not?	07
Q.3	(a) What is a semantic expansion?	03
	(b) List out advantages and disadvantages of macro.	04
	(c) Draw a flowchart and explain a simple one pass macro processor.	07
	OR	
Q.3	(a) What is a macro?	03
	(b) Explain recursive macro with example.	04
	(c) Explain design of macro preprocessor.	07
Q.4	(a) What is a macro preprocessor?	03
	(b) Define following terms: 1)START 2)END 3)BYTE 4)WORD	04
	(c) Explain the tasks performed by the PASS-1 and PASS-2 assembler?	07
	OR	
Q.4	(a) What is backtracking?	03
	(b) Define following terms: 1)OPTAB 2)SYMTAB 3)LITAB 4)POOLTAB	04

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| | (c) | Explain forward reference with suitable examples .How to solve it using back -patching? | 07 |
| Q.5 | (a) | Explain types of grammar. | 03 |
| | (b) | Differentiate Absolute loader and Direct linking loader. | 04 |
| | (c) | Explain any three Code Optimization Techniques. | 07 |
| | | OR | |
| Q.5 | (a) | Explain Ambiguous grammar with any suitable example. | 03 |
| | (b) | List out advantages of Relocating loader. | 04 |
| | (c) | Write Quadruple,Triples and Indirect Triples for expression, $x = -a*b + -a*b$ | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-VI (NEW) EXAMINATION – SUMMER 2022****Subject Code:3160715****Date:10/06/2022****Subject Name:System Software****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- Q.1**
- | | | |
|-----|---|-----------|
| (a) | Define system software. Give difference between system software and application software. | 03 |
| (b) | Explain the user-centric view and system-centric view of system software. | 04 |
| (c) | Explain lexical, syntax and semantic analysis with example. | 07 |
- Q.2**
- | | | |
|-----|--|-----------|
| (a) | Explain the following.
1. ORIGIN 2. EQU 3. LTORG | 03 |
| (b) | Show the difference between positional parameter and keyword parameter in macro. | 04 |
| (c) | Compare variant I and variant II of intermediate code. | 07 |
- OR**
- | | | |
|-----|---------------------------|-----------|
| (c) | Given the source program: | 07 |
|-----|---------------------------|-----------|

	START	200
	MOVER	AREG, ='5'
	MOVEM	AREG, M
L1	MOVER	AREG, ='2'
	ORIGIN	L1+3
	LTORG	
NEXT	ADD	AREG, ='1'
	SUB	BREG, ='2'
	BC	LT, BACK
	LTORG	
BACK	EQU L1	
	ORIGIN	NEXT+5
	MULT	CREG, ='4'
	STOP	
X	DS	1
	END	

1. Show the content of symbol table generated at the end of pass I.
2. Show the intermediate code generated for the program.

Q.3	(a) Compare and contrast the properties of macros and subroutines with respect to following criterion. 1. Code space requirement 2. Execution speed 3. Processing requirement by assembler 4. Flexibility	03
	(b) What is program relocation? How it is performed?	04
	(c) List and explain all the tables used in macro preprocessor.	07
OR		
Q.3	(a) Demonstrate the use of AIF and AGO.	03
	(b) Explain in brief about self relocating program.	04
	(c) List and explain all the task involved in macro expansion.	07
Q.4	(a) Explain compile-and-go loaders in brief.	03
	(b) What is debugger? Explain different types of error in program.	04
	(c) What is overlay? Explain the linking of overlay structured program performed.	07
OR		
Q.4	(a) Differentiate between linker and loader.	03
	(b) Differentiate pure and impure interpreter.	04
	(c) Write and explain an algorithm for first pass of the Linker program.	07
Q.5	(a) Explain Ambiguous Grammar.	03
	(b) Eliminate left recursion from the following grammar. $S \rightarrow Aa / b$ $A \rightarrow Ac / Sd / \epsilon$	04
	(c) What is optimizing transformation? discuss various optimizing transformations.	07
OR		
Q.5	(a) Define the following. 1. Finite state automaton 2. Regular expression 3. Operator grammar	03
	(b) Explain in brief about causes of large semantic gap.	04
	(c) Explain recursive descent parsing algorithm.	07
